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Analysis pipelines for calcium imaging data

Eftychios A Pnevmatikakis

Calcium imaging is a popular tool among neuroscientists because of its capability to monitor *in vivo* large neural populations across weeks with single neuron and single spike resolution. Before any downstream analysis, the data needs to be pre-processed to extract the location and activity of the neurons and processes in the observed field of view. The ever increasing size of calcium imaging datasets necessitates scalable analysis pipelines that are reproducible and fully automated. This review focuses on recent methods for addressing the pre-processing problems that arise in calcium imaging data analysis, and available software tools for high throughput analysis pipelines.

Address
Center for Computational Mathematics, Flatiron Institute, New York, NY 10010, United States

Corresponding author:
Pnevmatikakis, Eftychios A (epnevmatikakis@flatironinstitute.org)

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Introduction

Advances in optical microscopy techniques have resulted in datasets with rates that routinely reach or exceed 100 GB/h. This poses significant challenges in storage and all downstream processing. Any analysis pipeline needs to be fast, scalable to large datasets, and automated for efficiency and reproducibility reasons. The typical steps in a calcium imaging analysis pipeline are shown in Figure 1. The data in movie format is first motion corrected to remove artifacts arising from brain motion and slow imaging rate. Thereafter, the sources' (neurons, processes) locations and fluorescence traces are extracted from the motion corrected data. The fluorescence traces represent an indirect measure of neural activity can be further processed to infer the underlying activity in the form of spikes. Finally, an imaging experiment can visit the same field of view (FOV) in the brain over the course of multiple sessions/days. To combine the results from multiple sessions, the components from the different sessions need to be registered. Below we review recent approaches for all these problems, focusing primarily on the standard case of planar imaging, with cellular resolution, using either raster scanning two photon microscopy or one photon data through microendoscopic lenses.

Motion correction

Motion artifacts in calcium imaging datasets arise from natural brain movement. In small areas this motion can be approximated as rigid, and can be corrected using standard template based registration methods [58]. However brain motion can be faster than the raster scanning imaging rate, resulting in non uniform motion artifacts within a data frame, leading to non-rigid registration methods [17]. As the size of the imaged FOV grows brain motion itself within it can be non-rigid along all directions leading to methods that estimate motion in a piecewise-rigid manner [39]. Tracking methods [2] where the position of specific landmarks is tracked during an experiment can offer an alternative to template based methods. Such approached are used in whole brain imaging of freely moving animals where deformations can be highly complex; specialized tools have been developed for the case of C-elegans [32] or larval zebrafish [25,8]. While ground truth data for the motion correction problem is hard to obtain, several quality metrics have been proposed [39], and dual channel recordings where a structural indicator is also expressed can be used to evaluate the stability of various algorithms.

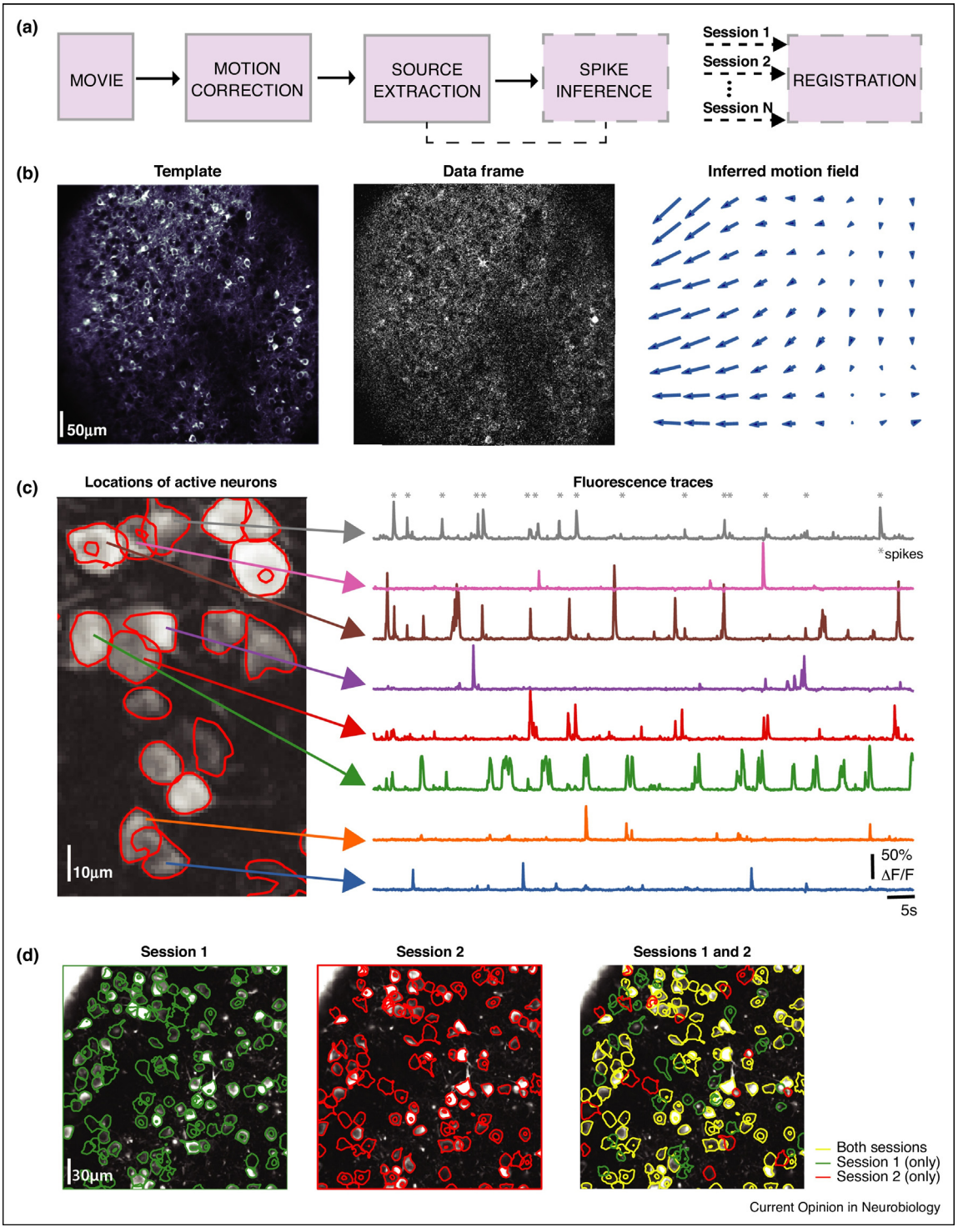
Source extraction

Detecting sources from summary statistics

A natural approach for detecting neurons in calcium imaging datasets, is by segmenting a static image that summarizes the observed data. Supervised learning methods using deep convolutional neural networks (CNN) have been used for this purpose. [4] proposes a CNN architecture for detecting neurons in max-projection images, whereas [26] uses a U-Net architecture for segmenting mean-activity images. These approaches require training on labeled datasets, and do not provide any temporal information. However, once applied, they can be used as initialization for full spatio-temporal approaches. Moreover, they can be used for detecting neurons that express the indicator but remain silent over the course of the imaging experiment. Unsupervised methods on static images, for example, dictionary learning [34] have also been used.

Summarizing datasets using a mean-projection or max-projection can result in densely packed images, since non-active expressing neurons are also present, which

Figure 1



Typical analysis pipeline for calcium imaging data. **(a)** The data is first processed for removing motion artifacts **(b)**. This can be done by estimating a motion field from aligning each data frame to a template. **(c)** Subsequently the locations of the neurons in the imaged FOV and their activity are extracted. Neurons can appear as spatially overlapping due to the limited axial resolution and their activity needs to be demixed. The activity of each neuron (as defined by the spikes depicted as gray stars) can be estimated from its corresponding fluorescence trace. **(d)** Registration of components produced by imaging sessions with the same FOV over the course of different days (*left, middle*). Neurons that are active in all or only some of the imaged sessions are identified (*right*). (The different steps of the pipeline are displayed on mouse *in vivo* cortex data, courtesy of S.A. Koay and D. Tank (Princeton University). Results were obtained using the CaImAn package [14*].)

can make the resulting segmentation problem particularly challenging. Alternative approaches use information about the similarities of different pixels across time. In an early approach, [49] used the correlation image to expose active sources that could then be segmented. Different methods create a matrix of distances between the traces of individual pixels, and apply graph cut [22,53] and spectral clustering methods [30•] to identify spatially connected regions corresponding to different active sources. Such approaches include no prior shape information and can be helpful in identifying non-somatic structures. Similar approaches based on grouping pixels with similar activity using active contour models have also been proposed [44].

Matrix factorization methods

A common approach for expressing the observed spatio-temporal fluorescence in terms of neural activity is by assuming that the fluorescence due to each active source can be expressed as a rank one matrix given by the outer product of two vectors: a vector in space $\boldsymbol{a}$ that represents the spatial footprint of each source in the FOV, and a vector in time $\boldsymbol{c}$ that represents its fluorescence activity. This approach gives rise to the following representation

$$\begin{aligned} F(x,t) &= \sum_{i=1}^{K} a_i(x)c_i(t) + B(x,t) \\ Y(x,t) &= F(x,t) + \varepsilon_{x,t}. \end{aligned} \tag{1}$$

Here, $F(\boldsymbol{x}, t)$ denotes the fluorescence at location $\boldsymbol{x}$ and time t. $\boldsymbol{a_i}$ and $\boldsymbol{c_i}$ denote, respectively, the spatial footprint and fluorescence trace of the component i, with $i = 1, \ldots, K$. $B(\boldsymbol{x}, t)$ denotes the neuropil/background activity at location $\boldsymbol{x}$ and time t. Finally, $Y(\boldsymbol{x}, t)$, $\varepsilon_{\boldsymbol{x},t}$ denote the observed fluorescence and measurement noise (assumed additive here for simplicity), respectively, at location $\boldsymbol{x}$ and time t. This representation has led to several source extraction approaches that aim to uncover it from the observed data, under various assumptions about the spatio-temporal structure of the components and the neuropil signal B.

Examples include extracting the pairs $(\boldsymbol{a_i}, \boldsymbol{c_i})$ using Independent Component Analysis (ICA) [31], or Non-negative Matrix Factorization (NMF) [29]. NMF objective functions with various constraints enforcing localized spatial footprints or certain dynamics in the extracted fluorescence traces, and different initialization methods, have led to algorithms that are used in practice for two photon [42•,18,19,55,35•] microendoscopic [63•], lightfield [33], or voltage [6] imaging data analysis. Matrix factorization can also be combined with dictionary learning methods [3,37], where the activity on each data frame is composed by a population of learned spatial footprints.

Neuropil removal

The recorded fluorescence during an imaging experiment typically contains a neuropil signal, which represents out of focus and scattered fluorescence from nearby processes (e.g. axons, dendrites, synaptic boutons). As a result a simple spatial weighted average of the data over the spatial footprint of a component can be contaminated by neuropil and not accurately represent its fluorescence trace. Neuropil contamination removal can be performed after the source detection step. [7] suggests drawing an annulus around each region of interest and use the average signal over that annulus for estimating the neuropil for this ROI. Similarly, [24] uses a NMF based approach around each ROI to separate the fluorescence trace from the neuropil. Neuropil removal can also be attempted prior to any pre-processing as shown in [27], where a morphological opening operator is used to estimate the background signal prior to motion correction.

An alternate approach is to explicitly model the spatio-temporal structure of the neuropil signal and estimate it as part of the source extraction problem (term $B(\boldsymbol{x}, t)$ in Eq. (1)). To account for long range correlations [42•] assumes a low rank spatio-temporal structure of the neuropil signal, and embed it within its CNMF framework. [14•] extends this approach by allowing the neuropil to have both localized and low rank structure. In [35•] $B(\boldsymbol{x}, t)$ is modeled by a linear combination of spatially localized basis functions. In the case of one-photon micro-endoscopic, the neuropil signal is much stronger due to the larger excitation volume and may also include the fluorescence of out of focus cell bodies. In the case of microendoscopic data, where the out of focus fluorescence is much stronger due to the larger integration volume, [63•] use an autoregressive model that models B at each location as the linear combination of the values of B in an annulus around it. The specific weights are learned from the algorithm.

Large scale analysis

Modern fluorescent microscopy methods in both one-photon and two-photon generate data at increasingly higher rates which make downstream processing (and even storing) challenging. Several solutions have been proposed to deal with this volume. [10] proposed an architecture for the distributed storage and processing of large datasets using a High Performance Computing (HPC) cluster. More flexible approaches that do not require access to a HPC have also been proposed. For example, in the CaImAn Batch algorithm [14•], the FOV is split in overlapping patches, that are processed in parallel with the CNMF algorithm [42•] and then appropriately combined. Data compression in the form of PCA [35•] or spatio-temporal decimation [11] has also been used as a pre-processing step to reduce the size of incoming data but also denoise it.

A more natural approach is by processing incoming data in an online streaming fashion. While methods for segmentation of individual static images [4,26] make a good candidate for this, they are sensitive to the noise that appears in individual data frames and do not readily give a framework for integrating information across time. A native approach comes with the `OnACID` algorithm [15•] that uses an online NMF framework for propagating the activity of identified sources, and tracks a rolling buffer of the unexplained variance to detect new sources as they appear, enabling the possibility of real time analysis and novel all-optical closed-loop experiments. In [14•] this approach was endowed with a CNN for identifying new neurons in the residuals, combining the benefits of online matrix factorization and segmentation of static images, to deal with large data rates.

Noise considerations

The source extraction problem is hindered by the presence of biological and measurement noise. Several approaches suggest using a denoising step as a first step prior to source extraction, such as PCA [31,35]. An approach more tailored to fluorescent imaging data which also achieves significant compression rates appears in Buchanan et al. [[56]]. Tepper et al. [56] estimates a scaled Poisson plus Gaussian noise model and appropriately transform the data to match the Gaussian noise assumptions present in many source extraction algorithms.

Benchmarking

Quantifying the performance of source extraction algorithms is a challenging task due to the scarcity of ground truth information. Ground truth can be derived through manual annotation and/or through the segmentation of datasets where the neurons co-express a static structural indicator typically localized to the nucleus. A collection of datasets with available ground is publicly available with the neurofinder challenge http://neurofinder.codeneuro.org/ and has been used for training algorithms and benchmarking different approaches using a precision recall metric. However, individual ground truth datasets can be unreliable: expression of structural indicators does not discriminate between active and non-active neurons, and is not guaranteed to be constrained only to neurons where the functional indicator is expressed. Moreover, as shown in Giovannucci et al. [14•], individual manual annotations can be highly variable with different labelers disagreeing up to a 20% level on the same dataset. Realistic generative models [51] can be valuable for this purpose, as well as creating labeled from multiple independent annotators [14•].

Spike inference

Spike inference methods aim to estimate the spike times of a neuron given the extracted fluorescence traces. This problem can be challenging due to the unknown and often non-linear nature of the indicator dynamics, the presence of measurement noise, and the relatively low imaging rate of typical recordings. A popular approach assumes linear and time invariant dynamics, expressing the fluorescence trace $\boldsymbol{c}$ (see Eq. (1)) as the sum of the calcium transients due to neural activity, plus measurement noise:

$$c(t) = \sum_{i=1}^{N} s_i h(t - t_i) + b + \varepsilon_t. \qquad (2)$$

Here t_i, s_i denote the time and amplitude of the transient i, $\boldsymbol{h}$ is a causal function denoting the shape of the transient, and b, ε_t denote the baseline (possibly time variant) and measurement noise at time t, respectively. Transients are typically characterized by fast rise followed by a slower decay. This behavior can be modeled by a simple single or double exponential model. Under these assumptions the deconvolution problem can be cast as a convex optimization problem that can be solved efficiently [60]. Several variants of this approach have also appeared proposing different objective functions based on sparsity assumptions and/or noise constraints [42•,59,21], while near-online formulations are also available [12•,20]. Spike inference methods based on detailed biophysical models have also been developed [16].

The approaches above estimate the amplitudes s_i of the calcium transients, a measure that is not directly interpretable in terms of number of spikes. Moreover, they provide a single estimate and do not quantify uncertainty. [13] bypasses this problem by integrating over the possible spike trains to provide estimates of underlying firing rates. Bayesian methods enable explicit estimation in the form of number of spikes [40,9•]. They can also provide uncertainty estimates for individual spikes, and in cases of high-SNR or multi-trial experiments provide estimates with resolution finer than the imaging rate [9•,38].

Supervised learning methods have also been applied to the spike inference problem [57]. These methods rely on training data, usually in the form of dual electrophysiological and imaging recordings in small neuron populations, which can be hard to obtain. A potential remedy comes from semi-supervised learning methods that also a learn a generative model [54•]. The recent community benchmark effort [5•] on the spikefinder challenge data (http://spikefinder.codeneuro.org/) revealed significant correlations between the results of supervised and unsupervised learning algorithms. The study used a correlation based metric; different metrics have also been proposed [57,45,36]. The theoretical limits of spike inference have also been studied [45,50] under the idealized linear conditions of Eq. (2).

Registration across multiple sessions

Calcium imaging enables the monitoring of large neural populations over many different sessions across multiple days. This gives rise to the problem of registering neurons across multiple sessions. Several packages offer semi-automated registration methods (e.g. [22,35•]). Fully automated approaches are also available [48•,14•] and they rely on the definition of a suitable metric for the distance between different spatial footprints across different sessions, and an alignment procedure based on the metric. Stability measures have also been proposed.

Software implementations

A critical requirement for the adoption of formal reproducible methods for analyzing calcium imaging data is the existence of reliable software that is scalable to the size of modern day datasets. Available packages for automated and/or interactive analysis include `SIMA` ([22] Python), `CaImAn` ([14•] (MATLAB and Python)), `Suite2p` ([35•] (MATLAB and Python)), `CNMF-E` ([63•] (MATLAB)), `ABLE` [44] (MATLAB)), `SCALPEL` ([37] (R)), `MIN1PIPE` ([27] (MATLAB)), `SamuROI` ([47] (Python)), and the toolbox of ([46] (MATLAB)).

Plain implementation of an algorithm is often not sufficient for the successful deployment, due to, for example, scalability issues and the vast diversity of imaging dataset produced in practice. Several of the software packages mentioned above, for example, `CaImAn` [14•], `CNMF-E` [63•], and `Suite2p` [35•], are under continual development beyond the respective algorithms they introduce. The goal is to provide generality, scalability and quality screening tests that aide towards analysis standardization and automation. The `CaImAn` package also provides a general infrastructure upon which different algorithms can be deployed [14•].

Current trends

Traditional two photon laser scanning microscopy is limited by the fact that every location has to be visited separately during each recording cycle. The experimenter faces a tradeoff between the size and resolution of the imaged FOV, and the acquisition frame rate. A popular choice for population recordings with single cell resolution uses a 512 × 512 pixel FOV with pixel size $\sim 1 \mu m^2$ obtained at a rate of around 30 Hz. Larger FOVs can be sampled at the expense of lower imaging rate, lower spatial resolution, or lower SNR and vice versa. A number of recent studies demonstrate that combining experimental design with data acquisition can improve this tradeoff by acquiring a low dimensional projection of the FOV, WF, where W is a, possibly time varying, measurement matrix, at faster rates or for a larger FOV F. Examples of such measurement designs can lead to several types of projections such as the sum of multiple individual planes [62], integration of larger volumes akin to spatial decimating [43,11], integration along axially elongated Bessel foci [28], V-shaped point spread functions [52], and tomographic measurements [23] (for a complete review see [61]). Even though motion correction can be challenging, especially when the projection measurement covers parts of the brain subject to different movement, these methods can in principle fit in a matrix factorization framework [41], and thus can be analyzed in a tractable way. In practice prior identification of the spatial footprints can significantly improve the source extraction problem [62,23], due to the high level of source mixing. While these methods can push the limits of fluorescent microscopy, their success is contingent on their successful deployment in multiple experimental labs, and specialized analysis methods can accelerate their adoption.

Discussion

The rapid growth in the adoption rate of fluorescence microscopy in systems neuroscience experiments has triggered the development of multiple data analysis methods, which rely on ideas from modern high dimensional statistics and machine learning. The increase in data acquisition rates has shown the necessity of scalable and reproducible analysis pipelines and their software implementations. Large collaborative experimental efforts such as the International Brain Laboratory [1], or the IARPA MICrONS project (https://www.iarpa.gov/index.php/research-programs/microns) are heavily dependent on high quality automated analysis tools, further demonstrating their exigency. We believe that online model based methods, endowed with supervised learning based neuron and event detection mechanisms can provide fast, scalable and interpretable solutions to analyzing calcium imaging data, and enable novel experimental design paradigms. Moreover, the methods developed for analyzing calcium imaging data are likely to find applications in other types of neuroscience applications such as voltage imaging and fluorescent reporting of neurotransmitter activity.

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An extension of the CNMF algorithm [42•] to microendoscopic imaging data that separates neural activity from the large amounts of out of focus fluorescence. Implementation is available as a standalone package in Matlab and as part of the CaImAn package [14•] in Python.